MH2000 : Experimental Methods
Thermodynamic Measurements
Dr. Sudhanshu Kuthe

Why do we perform thermodynamic experiments?

Thermodynamics predicts whether a reaction is favorable and which phases are stable at equilibrium. But predictions need reliable thermodynamic data. For real alloys and slags, components interact: their activities often differ from their mole fractions. Experiments allow us to measure equilibrium compositions, vapor pressures and phase behavior, then determine or validate activities, equilibrium constants and Gibbs-energy data used in process design.

A practical question: Will an alloying element remain dissolved in liquid metal, oxidize into slag, or evaporate? The answer depends on temperature, composition and chemical potentials not only on the nominal concentration.

From a measurable quantity to a thermodynamic property

$$\mu_i = \mu_i^\circ + RT \ln(a_i)$$

μ_i: chemical potential (J mol^{-1});
μ_i°: standard-state chemical potential;
R: gas constant;
T: temperature (K);
a_i: dimensionless activity.
For an alloy, $a_i = \gamma_i x_i$, where γ_i is the activity coefficient and x_i the mole fraction.

In the earlier transpiration experiment, a measured amount of vapor transported by a carrier gas can be used to infer equilibrium vapor pressure and, with a suitable reference state, activity. We often reverse the approach: impose a known gas chemical potential and observe how the sample responds at equilibrium.

Oxygen partial pressure versus oxygen potential

$$\mu O_2 = \mu O_2^\circ + RT \ln(pO_2/p^\circ)$$

The oxygen partial pressure pO_2 is a gas-phase pressure quantity; oxygen potential is the chemical potential of oxygen. At a fixed temperature they are linked by the equation above (ideal-gas approximation; p° is standard pressure). An extremely low equilibrium pO_2 can correspond to a strongly reducing atmosphere. In metallurgy, oxygen potential may alternatively be quoted per mole of O atoms: $\mu O = \frac{1}{2}\mu O_2$.

How CO-CO_2 gas mixtures control oxygen potential

$$2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$$

$$pO_2/p^\circ = [(pCO_2/pCO)^2] / K$$

At a fixed temperature, the CO_2/CO ratio sets an equilibrium oxygen potential through the reaction above (ideal-gas approximation). Increasing CO_2/CO raises oxygen potential.
With graphite present, the additional Boudouard equilibrium $C(gr) + CO_2\ (g) \rightleftharpoons 2CO(g)$ constrains the attainable gas composition; the imposed ratio cannot always be treated as independent of graphite equilibration.

Why it matters: a controlled oxygen potential lets us study oxide stability and equilibrium partitioning without attempting to meter extraordinarily small amounts of free oxygen gas.

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From controlled atmosphere to useful measurements

Consider a metal/oxide sample held at a known temperature in a controlled CO-CO_2 atmosphere. After sufficient equilibration, we determine the sample composition and phases. Together with the imposed oxygen potential, these observations can constrain oxide stability boundaries, oxygen activities and equilibrium constants. **Repeating the experiment at different temperatures and gas ratios gives thermodynamic data over a useful range**.

Other gas mixtures in this lecture

Gas mixture	Controlled quantity	Experimental purpose
H_2–H_2O	Oxygen potential	Set reducing/oxidizing conditions through $H_2 + ½O_2 \rightleftharpoons H_2O$.
H_2–H_2S	Sulfur potential	Establish sulfur chemical potential for sulfide equilibria.
Closed recirculating gas circuit	Gas composition/potential	Maintain a controlled atmosphere while circulating the gas.

Liquid-liquid chemical equilibria: metal and slag

In slag-metal or silicon-slag experiments, we measure how elements distribute between two liquid phases. At true equilibrium, the chemical potential of a transferable component is equal across the phases (with appropriate species and reaction constraints). Equilibrium concentrations can therefore provide distribution ratios and, with a suitable thermodynamic model or additional data, activities and reaction equilibrium constants. The lecture also shows mechanically induced emulsification. Stirring creates droplets and increases interfacial area, accelerating mass transfer. This is a kinetic effect: it helps the system approach equilibrium but does not by itself change the thermodynamic equilibrium state at fixed temperature, pressure and overall composition.

Thermodynamics and kinetics: do not confuse them

1. Thermodynamics: What equilibrium state is possible? What oxygen or sulfur potential stabilizes a phase?
2. Kinetics: How long does it take to reach equilibrium? How do mixing, diffusion and reaction rates affect the experiment?
3. Experimental caution: A measured composition is an equilibrium value only when equilibration has been demonstrated or adequately justified.

Three questions to discuss

1. Why can a CO-CO_2 mixture control oxygen potential even when almost no free O_2 is present?

2. If an oxide does not form immediately after changing the gas mixture, does that prove it is thermodynamically unstable? Why?

3. Why are measurements of final metal and slag compositions useful for developing thermodynamic databases?

We perform experiments to obtain and validate the thermodynamic data needed to predict real materials behavior. Controlled gas potentials establish known equilibrium conditions; measured phase compositions reveal how materials respond

MH2000 Experiment Simulator

Open the simulator: https://sudhanshukuthe.github.io/mh2000-simulator/
It runs in any web browser, on a laptop or a phone. No login, no installation.

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The experiments

#	Simulation	Direct link	What you can do
1	**Pseudoisopiestic method** (Au–Cd sealed tube, two-zone furnace)	#isopiestic	Set the reservoir and sample temperatures, wait for constant mass, and record points to build the a_{Cd}–x_{Cd} curve.
2	**Transpiration method** (vapour pressure of liquid Cu)	#transpiration	Run experiments at different carrier-gas flow rates and find the plateau between back-diffusion (low flow) and unsaturated gas (high flow). Compare Ar and He.
3	**C–CO–CO_2 equilibria** (controlled oxygen potential over graphite)	#boudouard	Change temperature and inlet gas, with or without graphite, and read the outlet %CO, p_{CO_2} and p_{O_2}.
4	**H_2–H_2O equilibrium** (water bubbler, controlled p_{O_2})	#hydrogen	Set the dew point and furnace temperature and see whether Fe, Ni or Cr oxidises or stays metallic.
5	**Liquid–liquid equilibria** (impeller mixing, induced emulsification)	#emulsion	Stir slag and liquid Si at different speeds and follow the interfacial area and the Ca transfer over time.

Each simulation has the animated apparatus, controls, live readouts and a chart of your results. Double-click the apparatus to show it full screen when projecting.

Suggested student exercises

1. **Pseudoisopiestic:** Record one point before equilibrium and one after. How far off is the early point, and why does the real Au-Mn study need 6–8 weeks?
2. **Transpiration:** Measure at 5, 20, 80 and 300 mL/min. Which flow rates give the true vapour pressure? What changes with He as the carrier gas?
3. **C–CO–CO_2:** Heat from 900 K to 1873 K with graphite present. Why can't an oxygen probe be used at 1873 K?
4. **H_2–H_2O:** For Fe at 1273 K, find the dew point at which FeO starts to form. Why does Cr oxidise even in very dry hydrogen?
5. **Slag-metal:** Compare 0, 50 and 100 rpm. Why does the interfacial area fall while mixing continues?

Scientific basis

The calculations use the same thermodynamic data as the lecture slides:

- $C(gr) + CO_2(g) = 2CO(g)$, $\Delta G^\circ = 170\,700 - 174.5T$ J
- $2CO(g) + O_2(g) = 2CO_2(g)$, $\Delta G^\circ = -564\,800 + 173.62T$ J
- $H_2(g) + \tfrac{1}{2}O_2(g) = H_2O(g)$, $\Delta G^\circ = -246\,450 + 54.8T$ J
- Cd and Cu vapour pressures from Clausius–Clapeyron fits; metal/oxide equilibria (Fe/FeO, Ni/NiO, Cr/Cr_2O_3) from standard compilations.

References

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